CS 7545 F26, Homework 3

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Exercise 1

Part A: Since $\mathbb{E}_z[z] = 0, \mathbb{E}_z\left[z^2\right] = 1$, we have the following equalities

$$L(\theta) = \mathop{\mathbb{E}}_{z}\left[(z-\theta)^2\right] + \lambda(\theta^2-1)^2 = \mathop{\mathbb{E}}_{z}\left[z^2 - 2z\theta\right] + \theta^2 + \lambda(\theta^2-1)^2 = 1 + \theta^2 + \lambda(\theta^2-1)^2;$$

$$\begin{aligned}\hat{L}_S(\theta) &= \frac{1}{n}\sum_{i=1}^{n}\left(z_i^2 - 2z_i\theta + \theta^2 + \lambda(\theta^2-1)^2\right) = \left(\frac{1}{n}\sum_{i=1}^{n} z_i^2\right) - 2\overline{z}_S\theta + \theta^2 + \lambda(\theta^2-1)^2 \\ &= \left(\frac{1}{n}\sum_{i=1}^{n} z_i^2\right) + (\theta - \overline{z}_S)^2 - \overline{z}_S^2 + \lambda(\theta^2-1)^2\end{aligned}$$

Part B:
1. With $\lambda = \frac{1}{2}$, $L(\theta) = \frac{1}{2}(\theta^4+3)$ and $L''(\theta) = 6\theta^2$. Therefore, the minimizer $\theta^\star = 0$ and $L''(\theta^\star) = 0$.
2. $\hat{L}'_S(\theta) = 2(\theta^3 - \overline{z}_S)$. Since $\hat{L}'_S(\theta)$ is strictly increasing, the ERM is unique and $\hat{L}'_S(\hat{\theta}_S) = 0$ implies that $\hat{\theta}_S = (\overline{z}_S)^{1/3}$.
3. We have that $L(\theta^\star) = 3/2$ and

$$\mathbb{E}\left[L(\hat{\theta}_S) - L(\theta^\star)\right] = \mathbb{E}\left[1 + (\overline{z}_S)^{2/3} + \frac{1}{2}\left((\overline{z}_S)^{2/3} - 1\right)^2\right] - \frac{3}{2} = \frac{1}{2}\mathbb{E}\left[(\overline{z}_S)^{4/3}\right] = \frac{1}{2}\mathbb{E}\left[|\overline{z}_S|^{4/3}\right].$$

Since $x \mapsto x^{4/3}$ is convex, by Jensen's inequality, $\mathbb{E}\left[L(\hat{\theta}_S) - L(\theta^\star)\right] = \frac{1}{2}\mathbb{E}\left[|\overline{z}_S|^{4/3}\right] \geq \frac{1}{2}\mathbb{E}\left[|\overline{z}_S|\right]^{4/3}$.
4. Since $z_i \sim \mathcal{N}(0,1)$, $\overline{z}_S \sim \mathcal{N}(0, 1/n)$. Therefore, the expected excess loss satisfies that

$$\mathbb{E}\left[L(\hat{\theta}_S) - L(\theta^\star)\right] \geq \frac{1}{2}\mathbb{E}\left[|\overline{z}_S|\right]^{4/3} = \frac{1}{2}\left(\sqrt{\frac{2}{\pi n}}\right)^{4/3} = \Omega\left(n^{-2/3}\right).$$

Part C: 1. $L(\theta) = \frac{2}{3}(\theta^2 - \frac{1}{4})^2 + \frac{13}{8}$. Therefore, the global minimizers satisfy $(\theta^\star)^2 = \frac{1}{4}$, and hence, $\theta^\star = \pm\frac{1}{2}$.
2. Direct calculation shows that $\hat{L}_S(\theta) - \hat{L}_S(-\theta) = (\theta - \overline{z}_S)^2 - (\theta + \overline{z}_S)^2 = -4\overline{z}_S\theta$. Since $\hat{\theta}_S$ is the minimizer, $\hat{L}_S(-\hat{\theta}_S) - \hat{L}_S(\hat{\theta}_S) = 4\overline{z}_S\hat{\theta}_S \geq 0$. Also, $\hat{L}'_S(0) = -2\overline{z}_S \neq 0$, implying $\hat{\theta}_S \neq 0$. Therefore, $\hat{\theta}_S$ has the same sign as $\overline{z}_S$.
3. $\hat{L}'_S(\theta) = \frac{8}{3}\theta^3 - \frac{2}{3}\theta - 2\overline{z}_S$. Since $\hat{\theta}_S$ is the minimizer, $\hat{L}'_S(\hat{\theta}_S) = 0$, that is, $4\hat{\theta}_S^3 - \hat{\theta}_S = 3\overline{z}_S$. If $\overline{z}_S$ is positive, then $\mathrm{sgn}(\hat{\theta}_S) > 0$ and $\hat{\theta}_S \geq \frac{1}{2}$. In this case, $3\overline{z}_S = (2\hat{\theta}_S + 1)(2\hat{\theta}_S - 1)\hat{\theta}_S \geq 2(\hat{\theta}_S - \frac{1}{2})$. If $\overline{z}_S$ is negative, then $\mathrm{sgn}(\hat{\theta}_S) < 0$ and $\hat{\theta}_S \leq -\frac{1}{2}$. In this case, $3\overline{z}_S = (2\hat{\theta}_S + 1)(2\hat{\theta}_S - 1)\hat{\theta}_S \leq 2(\hat{\theta}_S + \frac{1}{2})$.

Hence, $\left|\hat{\theta}_S - \frac{1}{2}\mathrm{sgn}(\overline{z}_S)\right| \le \frac{3}{2}\left|\overline{z}_S\right|$.

4. By Gaussian tail bound, $\Pr\left[|\overline{z}_S| \ge t\right] \le 2\exp(-nt^2/2)$. By $\left|\hat{\theta}_S - \frac{1}{2}\mathrm{sgn}(\overline{z}_S)\right| \le \frac{3}{2}\left|\overline{z}_S\right|$, we conclude that

$$\Pr\left[\left|\hat{\theta}_S - \frac{1}{2}\mathrm{sgn}(\overline{z}_S)\right| \le \frac{3}{2}t\right] \ge \Pr\left[|\overline{z}_S| \le t\right] \ge 1 - 2\exp\left(-nt^2/2\right).$$

By taking $t = \sqrt{2\ln(2/\delta)/n}$, we complete the proof.

Exercise 2

We have that $d(x,x) = \sqrt{2k(x,x) - 2k(x,x)} = 0$. The symmetry of d follows from the symmetry of k. Define the matrix $K \in \mathbb{R}^{\mathcal{X}\times\mathcal{X}}$ and $K = (k(x,y))_{x,y\in\mathcal{X}}$ is PSD. Define $\mathbf{1}_{a,b}$ to be the vector satisfying that $\mathbf{1}_{a,b}(x) = \mathbb{1}[x = a] - \mathbb{1}[x = b]$. Then $d(x,y) = \sqrt{\mathbf{1}_{x,y}^\top K \mathbf{1}_{x,y}}$. For the triangle inequality, we have that

$$\begin{aligned}
d(x,y) &= \sqrt{\mathbf{1}_{x,y}^\top K \mathbf{1}_{x,y}} = \sqrt{(\mathbf{1}_{x,z} + \mathbf{1}_{z,y})^\top K (\mathbf{1}_{x,z} + \mathbf{1}_{z,y})} \\
&= \sqrt{\mathbf{1}_{x,z}^\top K \mathbf{1}_{x,z} + \mathbf{1}_{z,y}^\top K \mathbf{1}_{z,y} + 2\left\langle K^{1/2}\mathbf{1}_{x,z}, K^{1/2}\mathbf{1}_{z,y}\right\rangle} \\
&\le \sqrt{\left\|K^{1/2}\mathbf{1}_{x,z}\right\|_2^2 + \left\|K^{1/2}\mathbf{1}_{z,y}\right\|_2^2 + 2\left\|K^{1/2}\mathbf{1}_{x,z}\right\|_2\left\|K^{1/2}\mathbf{1}_{z,y}\right\|_2} \\
&= d(x,z) + d(z,y),
\end{aligned}$$

where the inequality follows from the Cauchy-Schwarz inequality.

Exercise 3

Let $K = (k(x_i,x_j))_{i,j\in[n]}$. Then by Section 3.4 in Lecture notes 9, the prediction on a test point x_0 is

$$k_{x_0}^\top (K + n\lambda I)^{-1} y \qquad \text{where} \qquad (k_{x_0})_i = k(x_i, x_0). \tag{1}$$

Define $C_1, \ldots, C_p$ to be the "clusters" of the training set which satisfies that for any $i, j \in [p]$ and $a \in C_i, b \in C_j$, $i = j \iff k(x_a, x_b) = 1$. We use $\mathbf{1}_S$ to denote the indicator vector, that is, $\mathbf{1}_S(x) = \mathbb{1}[x \in S]$. Therefore, by definition, $K = \sum_{i=1}^p \mathbf{1}_{C_i}\mathbf{1}_{C_i}^\top$. From the Sherman–Morrison formula, we can observe that the inverse of $K + n\lambda I$ should have the following form,

$$\left(n\lambda I + \sum_{i=1}^p \mathbf{1}_{C_i}\mathbf{1}_{C_i}^\top\right)^{-1} = (n\lambda)^{-1} I + \sum_{i=1}^p \alpha_i \mathbf{1}_{C_i}\mathbf{1}_{C_i}^\top,$$

where $\{\alpha_i\}_{i\in[p]}$ are coefficients that need to be determined. Since $\left\langle \mathbf{1}_{C_i}, \mathbf{1}_{C_j}\right\rangle = 0$ for $i \ne j$, direct calculation shows that

$$\left((n\lambda)^{-1} I + \sum_{i=1}^p \alpha_i \mathbf{1}_{C_i}\mathbf{1}_{C_i}^\top\right)\left(n\lambda I + \sum_{i=1}^p \mathbf{1}_{C_i}\mathbf{1}_{C_i}^\top\right) = I + \sum_{i=1}^p ((n\lambda)^{-1} + n\alpha_i\lambda + |C_i|\alpha_i)\mathbf{1}_{C_i}\mathbf{1}_{C_i}^\top.$$

Setting $\alpha_i = -(n\lambda)^{-1}(n\lambda + |C_i|)^{-1}$, we get the inverse of $K + n\lambda I$.

If $k(x_0, x_i) = 0$ for some $i \in [n]$, then by Equation (1), $k_{x_0} = 0$ and the prediction is 0, as desired.

Suppose $k(x_0, x_i) = 1$ for some x_i and $i \in C_\ell$. We have that by the definition of the partition, $C(x_0) = C_\ell$. Therefore, by Equation (1), the prediction is as follows,

$$\begin{aligned}
\mathbf{1}_{C_\ell}^\top \left((n\lambda)^{-1} I - \sum_{i=1}^{p} \frac{1}{n\lambda(n\lambda + |C_i|)} \mathbf{1}_{C_i} \mathbf{1}_{C_i}^\top \right) y &= \mathbf{1}_{C_\ell}^\top \left((n\lambda)^{-1} I - \frac{1}{n\lambda(n\lambda + |C_\ell|)} \mathbf{1}_{C_\ell} \mathbf{1}_{C_\ell}^\top \right) y \\
&= \left((n\lambda)^{-1} - \frac{|C_\ell|}{n\lambda(n\lambda + |C_\ell|)} \right) \mathbf{1}_{C_\ell}^\top y \\
&= \frac{\sum_{i \in C(x_0)} y_i}{n\lambda + |C(x_0)|}.
\end{aligned}$$

Exercise 4

1. Conditioned on $h^\star$, we have that

$$\Pr\left[\hat{L}_S(h) = 0\right] = \Pr\left[\forall i \in [n], h(x_i) = h^\star(x_i)\right] = (1 - \Pr_{x \sim \mathcal{D}}\left[h(x) \neq h^\star(x)\right])^n \leq \exp(-n L_{h^\star}(h)). \quad (2)$$

2. Fix a hypothesis h and $h^\star$. If $L_{h^\star}(h) \leq \frac{\ln(1/\pi(h)) + \ln(1/\delta)}{n}$, the event in the problem description holds with probability 1. Otherwise by Equation (2), the following holds,

$$\Pr\left[\hat{L}_S(h) = 0 \mid h^\star\right] \leq \exp(-n L_{h^\star}(h)) \leq \exp(-(\ln(1/\pi(h)) + \ln(1/\delta))) = \pi(h)\delta. \quad (3)$$

Taking the complementary event completes the proof.

3. Applying the union bound over all h in Equation (3), we have that

$$\Pr\left[\exists h, \hat{L}_S(h) = 0 \text{ and } L_{h^\star}(h) > \frac{\ln(1/\pi(h)) + \ln(1/\delta)}{n} \mid h^\star\right] \leq \delta$$

Note that for the posterior distribution, we have that $\Pr\left[h = h^\star \mid S\right] \propto \pi(h)\mathbb{1}[\hat{L}_S(h) = 0]$, thus, $\hat{L}_S(\hat{h}_{\text{MAP}}(S)) = 0$. Therefore,

$$\begin{aligned}
&\Pr\left[L_{h^\star}\left(\hat{h}_{\text{MAP}}(S)\right) \leq \frac{\ln(1/\pi(\hat{h}_{\text{MAP}}(S))) + \ln(1/\delta)}{n} \mid h^\star\right] \\
\geq &\Pr\left[\forall h, \hat{L}_S(h) > 0 \text{ or } L_{h^\star}(h) \leq \frac{\ln(1/\pi(h)) + \ln(1/\delta)}{n} \mid h^\star\right] \geq 1 - \delta. \quad (4)
\end{aligned}$$

4. By the maximality of $\hat{h}_{\text{MAP}}(S)$, $\pi(\hat{h}_{\text{MAP}}(S)) \geq \pi(h^\star)$ holds. By Equation (2), we conclude that

$$\begin{aligned}
\Pr\left[L_{h^\star}\left(\hat{h}_{\text{MAP}}(S)\right) \geq \varepsilon \mid h^\star\right] &\leq \Pr\left[\exists h, \hat{L}_S(h) = 0 \text{ and } L_{h^\star}(h) \geq \varepsilon \text{ and } \pi(h) \geq \pi(h^\star) \mid h^\star\right] \\
&\leq \sum_{\substack{L_{h^\star}(h) \geq \varepsilon \\ \pi(h) \geq \pi(h^\star)}} \Pr\left[\hat{L}_S(h) = 0 \mid h^\star\right] \\
&\leq e^{-n\varepsilon} \sum_{h : \pi(h) \geq \pi(h^\star)} \frac{\pi(h)}{\pi(h)} \leq e^{-n\varepsilon} \sum_{h : \pi(h) \geq \pi(h^\star)} \frac{\pi(h)}{\pi(h^\star)} \leq \frac{e^{-n\varepsilon}}{\pi(h^\star)}, \quad (5)
\end{aligned}$$

where the second inequality follows from the union bound.
5. Fix $T = \ln(1/\pi(h^\star))/n$. We have that

$$\begin{aligned}\mathop{\mathbb{E}}_{S}\left[L_{h^\star}\left(\hat{h}_{\mathrm{MAP}}(S)\right) \mid h^\star\right] &= \int_0^\infty \Pr\left[L_{h^\star}\left(\hat{h}_{\mathrm{MAP}}(S)\right) > t \mid h^\star\right] \mathrm{d}t \leq T + \int_T^\infty \frac{e^{-nt}}{\pi(h^\star)}\,\mathrm{d}t \\ &= \frac{1+\ln(1/\pi(h^\star))}{n}. \end{aligned} \tag{6}$$

6. By Equation (6), we conclude that

$$\begin{aligned}\Pr_{S,h^\star}\left[\hat{h}_{\mathrm{MAP}}(S)(x_0) \neq y_0\right] &= \mathop{\mathbb{E}}_{S,h^\star,x_0}\left[\mathbb{1}\left[\hat{h}_{\mathrm{MAP}}(S)(x_0) \neq h^\star(x_0)\right]\right] = \mathop{\mathbb{E}}_{S,h^\star}\left[L_{h^\star}\left(\hat{h}_{\mathrm{MAP}}(S)\right)\right] \\ &= \mathop{\mathbb{E}}_{S,h^\star}\left[\mathbb{E}\left[L_{h^\star}\left(\hat{h}_{\mathrm{MAP}}(S)\right) \mid h^\star\right]\right] \leq \frac{1+\mathbb{E}_{h^\star}\left[\ln(1/\pi(h^\star))\right]}{n} \\ &= \frac{1+\mathrm{Entropy}[\pi]}{n}.\end{aligned}$$

Exercise 5

1. Direct calculation gives

$$\Pr\left[\theta^\star = \theta \mid \{(x_i,y_i)\}_{i=1}^n\right] \propto \exp\left(-\frac{\|\theta\|_2^2}{2\lambda}\right) \times \exp\left(-\sum_{i=1}^n \frac{(\theta\cdot x_i - y_i)^2}{2\sigma^2}\right).$$

Therefore, $E(\theta) = \frac{\|\theta\|_2^2}{2\lambda} + \sum_{i=1}^n (\theta\cdot x_i - y_i)^2/(2\sigma^2)$.
2. We write $X = (x_1,\ldots,x_n)$, $Y = (y_1,\ldots,y_n)^\top$ and $\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^n x_i x_i^\top$. The derivatives of E are as follows,

$$\nabla E(\theta) = \frac{\theta}{\lambda} + \frac{XX^\top\theta - XY}{\sigma^2} \qquad \text{and} \qquad \nabla^2 E(\theta) = \frac{I_d}{\lambda} + \frac{n}{\sigma^2}\hat{\Sigma}.$$

Since $\hat{\theta}_{\mathrm{MAP}}$ minimizes E, $\nabla E(\theta) = 0$ implies

$$\hat{\theta}_{\mathrm{MAP}} = \left(\frac{\sigma^2 I_d}{\lambda} + XX^\top\right)^{-1} XY.$$

Therefore, the Laplace approximation is as follows,

$$\Pr\left[\theta^\star = \theta \mid \{(x_i,y_i)\}_{i=1}^n\right] \propto \exp\left(-\frac{1}{2}(\theta-\hat{\theta}_{\mathrm{MAP}})^\top\left(\frac{I_d}{\lambda} + \frac{n}{\sigma^2}\hat{\Sigma}\right)(\theta-\hat{\theta}_{\mathrm{MAP}})\right).$$

In this setting, the mean and covariance are as follows,

$$\mu = \left(\frac{\sigma^2 I_d}{\lambda} + XX^\top\right)^{-1} XY, \qquad \Sigma = \left(\frac{I_d}{\lambda} + \frac{n}{\sigma^2}\hat{\Sigma}\right)^{-1}.$$

3. Direct calculation gives

$$\Pr\left[\theta^\star = \theta \mid \{(x_i,y_i)\}_{i=1}^n\right] \propto \exp\left(-\frac{\|\theta\|_2^2}{2\lambda}\right) \times \prod_{i=1}^n \sigma\left(y_i\left\langle\theta, x_i\right\rangle\right).$$

Therefore, $E(\theta) = \frac{\|\theta\|_2^2}{2\lambda} - \sum_{i=1}^n \ln(\sigma(y_i \langle \theta, x_i \rangle))$.
4. The derivatives of E are as follows,

$$\nabla E(\theta) = \frac{\theta}{\lambda} - \sum_{i=1}^n y_i \sigma(-y_i \langle x_i, \theta \rangle) x_i \qquad \text{and} \qquad \nabla^2 E(\theta) = \frac{I_d}{\lambda} + \sum_{i=1}^n \sigma(\langle x_i, \theta \rangle) \sigma(-\langle x_i, \theta \rangle) x_i x_i^\top.$$

The maximizer $\hat{\theta}_{\text{MAP}}$ satisfies $\nabla E(\hat{\theta}_{\text{MAP}}) = 0$. For the Laplace approximation, we have that the posterior distribution is proportional to

$$\exp\left(-\frac{1}{2}(\theta - \hat{\theta}_{\text{MAP}})^\top \left(\frac{I_d}{\lambda} + \sum_{i=1}^n \sigma\left(\left\langle x_i, \hat{\theta}_{\text{MAP}} \right\rangle\right) \sigma\left(-\left\langle x_i, \hat{\theta}_{\text{MAP}} \right\rangle\right) x_i x_i^\top\right) (\theta - \hat{\theta}_{\text{MAP}})\right).$$

In this setting, the mean and covariance are as follows,

$$\mu = \hat{\theta}_{\text{MAP}}, \qquad \Sigma = \left(\frac{I_d}{\lambda} + \sum_{i=1}^n \sigma\left(\left\langle x_i, \hat{\theta}_{\text{MAP}} \right\rangle\right) \sigma\left(-\left\langle x_i, \hat{\theta}_{\text{MAP}} \right\rangle\right) x_i x_i^\top\right)^{-1}.$$